

Experimental and numerical investigation of direct liquid jet impinging cooling using 3D printed manifolds on lidded and lidless packages for 2.5D integrated systems



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HIGHLIGHTS

- A scalable impingement cooling solution for multi-chip packages is proposed.
- A novel 3D printed lateral delivery manifold with low pressure drop is presented.
- Experimental performance benchmarking for lidded and lidless packages is performed.
- Very low inter-chip thermal coupling (4%) is measured for the bare die cooling.

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ABSTRACT

Package level bare die jet impingement cooling on the chip backside has been previously demonstrated as a highly efficient jet cooling solution for high power devices, by eliminating the thermal interface material. However, bare die jet impingement cooling is a disruptive cooling technology requiring direct access to the backside of the semiconductor device. A less disruptive cooling implementation in which the impingement cooling is applied on the lid is now investigated as an intermediate step. In this work, the first comparison of the cooling performance with the liquid micro-jet array impingement cooling on lidded and bare die package is conducted and experimentally characterized for polymer coolers fabricated by additive manufacturing. The thermal characterization is performed on a 2.5D integrated system that contains two advanced thermal test dies. In addition, besides the reference cooler with vertical coolant supply channel, an improved design of 3D printed fluid delivery manifold with lateral inlet/outlet channels is introduced and benchmarked. The experimental and numerical comparison show that the cooler with the lateral feeding manifold requires 60% less pumping power with respect to the vertical feeding manifold for the same high thermal performance while reducing the overall thickness of the cooling solution by a factor of 2.

1. Introduction

1.1. Need for advanced cooling solutions for 2.5D Si interposer packages

Three-dimensional through-Si via (TSV) integration has great potential to improve the performance, power consumption, and package footprint by vertically integrating multiple dies [1]. However, the vertical integration with 3D stacked dies will elevate the power density and chip temperature, which requires expensive packaging and cooling solutions [2]. This is due to the thermal bottleneck of the die-die interface materials with low thermal conductivities [3]. Alternatively, 2.5D Si interposer packages with multiple dies integrated side by side,

enable more cooling potential for applications combining high power components such as logic, GPU and FPGA, and temperature sensitive components (DRAM, SerDes). This Si interposer implementation shows potential for high performance systems with high-bandwidth and high-power applications [4], as demonstrated by the release of Xilinx's FPGA [5] and the AMD Fury X GPU [6].

For typical 2.5D Si interposer packages, a metal lid or heat spreader is attached on the substrate using lid adhesive [7,8]. In literature [7], several thermal solutions are mounted on top of the lidded package to minimize the thermal resistance, such as fin heat sinks, and fan cooled heat sink with or without embedded heat pipes. However, the major thermal bottleneck for conventional cooling solutions is the presence of

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the thermal interface material (TIM). The thermal resistivity of the most widely used TIMs such as greases, gels, and phase-change materials (PCMs), can be as low as $10 \text{ mm}^2\text{-K/W}$ [9]. For the state of art nano-TIM, the thermal resistivity can be lower and even in the range of $1 \text{ mm}^2\text{-K/W}$ with GE's copper nanospring [10]. However, it is also found that the interfacial thermal resistance (ITR) between TIM and heat sink can vary from $2 \text{ mm}^2\text{-K/W}$ to $20 \text{ mm}^2\text{-K/W}$ due to the mechanical compliance of the TIM [11]. Recently, several embedded cooling techniques without the use of the TIM have been applied on the 2.5D Si interposer packages. In [12] an embedded thermoelectric cooler (TEC) combined with silicon interposer for the electrical path is studied for hot spot cooling, but the power consumption of the TEC driver is a big challenge. In [13–15], microfluidic cooling delivery channels are embedded within an interposer package with high aspect ratio TSVs, and microfluidic chip I/Os. However, the I/O density is insufficient for high-bandwidth devices. Moreover, the temperature gradient across and along the channels is hard to avoid, resulting in an uneven temperature distribution.

1.2. Overview of the 3D distributed manifold techniques

Impingement jet cooling is a potential solution for the high-performance system. Over the last decades, the heat transfer and flow dynamics physics and correlations of jet impingement cooling with electronics and modules were investigated systematically [16–19]. As shown in Fig. 1, a scalable multiple impingement jet cooling system with liquid distributed manifold is applied directly on the die surface. For the microchannel cooling heat sink, it is shown that the design of manifold microchannel (MMC) heat sink with alternating inlet and outlet channels has big impact on the system pressure drop and thermal performance [20,21]. The experimental study shows that the thermal resistance of the MMC heat sink is approximately 35% lower than the traditional microchannel heat sink [21]. In literature [22], numerical simulation results indicate that an optimized manifold design can reduce the thermal resistance by 50% compared to a traditional microchannel heat sink. Moreover, a numerical study by Boteler et al. [23] indicate a more uniform flow distribution and lower pressure drop by as much as 97% for a 3D manifold design compared to a traditional microchannel design. Therefore, the manifold level design of this microfluidic cooler is very important for the overall cooler performance, since it determines the flow uniformity, and system level pressure and thermal resistance, especially for large area die size applications.

For the manufacturing of the impingement jet cooling system with 3D distribution manifold, different fabrication techniques were investigated, such as plastic [24], metal micromachining [25] and Si deep reactive-ion etching (DRIE) microfabrication [26], multilayer ceramic technology (MLC) [27]. However, these fabrication techniques are all very expensive. In our previous research, a high efficiency polymer

cooler has been demonstrated using mechanical micromachining process [28]. Fig. 1 shows the schematic of this cooling concept. However, the different structural layers shown in Fig. 1(a) have to be fabricated separately, increasing the reliability concerns of the assembly.

With the recent development of the high resolution of additive manufacturing technology, lots of studies move to the 3D manifold with liquid delivery system, fabricated by 3D printing. Robinson et al., demonstrated a hybrid micro heat sink using impinging micro-jet arrays and microchannels using MICA Freeform process [29], with a predicted effective thermal conductance of $400 \text{ kW/m}^2 \text{ K}$ for a flow rate of 0.5 L/min . In [30], we demonstrated for a bare die single chip package that additive manufacturing, or 3D printing can be used to fabricate a highly performant cooler with high density nozzle array and complex internal geometries. Moreover, 3D printing of the cooler shows the potential to integrate the cooling jets directly targeted on each device in the multi-chip module, which can drastically reduce the thermal coupling between different devices. Furthermore, the required pumping power can be significantly reduced due to a streamlined internal channel design which can be fabricated using additive manufacturing, but not with conventional fabrication techniques.

In this paper, we present the evaluation of the cooling performance of a package level jet impingement cooling solution on dual-chip packages by means of full cooler computational fluid dynamics simulations (CFD) and experiments on test vehicles. For this study the single chip cooler design [30] has been optimized for the packages containing two thermal test chips. Furthermore, an improved cooler design of the 3D printed fluid manifold is introduced and benchmarked. Since jet impingement cooling on the bare die is a disruptive cooling technology requiring direct access to the backside of the Si chip, we also consider a less disruptive cooling implementation in which the impingement cooling is applied on the lid. For cooling on the lidded package, the cooling surface area of the lid is larger compared to cooling on the bare die surface. Moreover, the presence of the lid enhances the lateral heat spreading effects inside the package. However, the drawback of this lidded approach is the significant thermal resistance of the TIM between the chip and the lid. The trade-off between the beneficial and detrimental effect of the lid, will determine whether the bare die cooling will outperform the cooling on the lidded package.

The paper is organized as follows: the cooler design considerations are discussed in Section 2. The fabricated cooler, the assembly and the experimental set-up for the thermal and hydraulic characterization are presented in Section 3. In Section 4, the experimental characterization of the package level impingement cooler is analyzed for the bare die and lidded package configurations, including the thermal coupling between the dies in the package. Finally, Section 5 discusses the experimental and numerical characterization of the novel cooler design with smaller form factor and the comparison with the reference cooler.

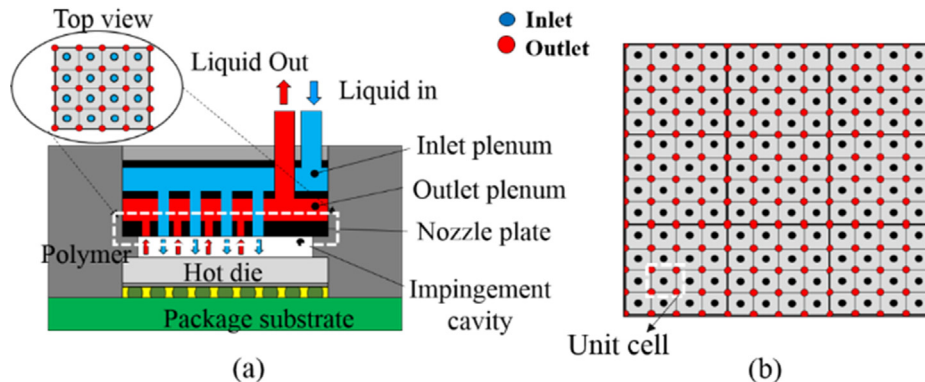


Fig. 1. Scalable impingement jet cooling system with 3D liquid distributed manifold: (a) cross section view of the cooler on single chip; (b) top view of the nozzle plate with scalable nozzle array.

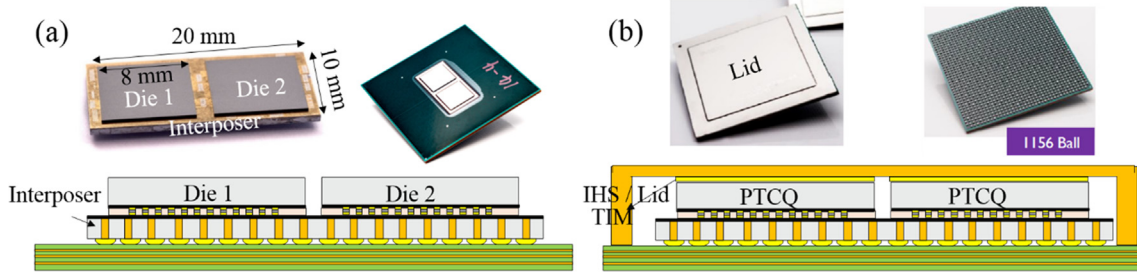


Fig. 2. Bare die package and lidded package: (a) two thermal test chips on Si interposer without lid; (b) thermal test chips on Si interposer with lid.

2. Design of multi-jet impingement cooler

This section introduces the design considerations for the impingement cooler of the dual-chip package. Firstly, the lidded and bare die thermal test vehicles on 2.5D interposer are introduced. Secondly, the concept of nozzle array scalability with the chip area is introduced in order to estimate the cooler performance based on the extrapolation of previous cooler designs. Next, the different design concepts for the package level impingement cooler are discussed. Finally, the package level CFD modeling approach is presented to study the flow and temperature distribution inside the cooler and the overall cooler performance in detail for the different proposed cooler designs.

2.1. Lidded and bare die dual-chip test vehicle

In order to compare the performance between the bare die cooling and lidded package cooling, an advanced thermal test vehicle with lidded package and bare die package versions is introduced. As shown in Fig. 2, a $35 \times 35 \text{ mm}^2$ ball grid array (BGA) package is used, containing a $20 \times 10 \text{ mm}^2$ Si interposer with $100 \mu\text{m}$ thickness and two identically $8 \times 8 \text{ mm}^2$ thermal test chips, referred to as PTCQ (Packaging Test Chip version Q) [31,32]. The interposer stacks are flip-chip soldered on the organic substrate, allowing the cooling solution to be directly applied to the backside of the chips, or on the lid. The schematic of the bare die package is illustrated in Fig. 2(a). As for the lidded packages shown in Fig. 2(b), a Cu lid with a thickness of 0.3 mm is attached on the thermal test dies with additional thermal interface material. The thermal interface material is a standard silicone based material with a specified thermal conductivity of 1.9 W/m K and a targeted thickness of $80 \mu\text{m}$.

The details of the $8 \times 8 \text{ mm}^2$ PTCQ thermal test chip are shown in Fig. 3. The dedicated thermal test chip with an array of 32×32 individually controllable cells can be programmed to generate a custom power map, ranging from localized hot spots to distribution with quasi-uniform heating, with 75% area uniformity shown in Fig. 3(b). The full temperature map can be measured with a 32×32 array of diodes across the $8 \times 8 \text{ mm}^2$ area of the chip. The voltage drop across the

diode for a constant current is used as the temperature sensitive parameter of the sensor. The 95% confidence interval of the calibrated sensitivity is $-1.55 \pm 0.02 \text{ mV}/^\circ\text{C}$ for a current of $5 \mu\text{A}$ in the temperature range between 10 and 75°C . The chip temperature sensors allow absolute temperature measurements with an accuracy of $\pm 2\text{--}3^\circ\text{C}$. The test chip can generate a maximum power 150 W/cm^2 in each heater cell or 80 W in the whole test chip for all heater cells activated. The layout of the heater cells is shown in Fig. 3(c).

2.2. Nozzle array scalability

As shown in Fig. 1, the nozzle array pattern is a scalable system of repeated unit cells with a centered inlet and outlets in the corners serving for the neighboring cells too, under the assumption of identical thermal performance for all unit cells. The nozzle array of unit cells scales with the chip area. Therefore, the thermal and hydraulic performance metrics can be reported as area independent thermal resistance and pump power to represent the intrinsic thermal and hydraulic behavior. The overall thermal performance of the cooler is expressed in terms of the thermal resistance R_{th} and the pump power W_p .

$$R_{th} = \frac{T_{avg} - T_{in}}{Q_{heater}} \quad (1)$$

$$W_p = \dot{V} \cdot \Delta P \quad (2)$$

where T_{avg} is the measured average chip temperature, T_{in} is the coolant inlet temperature, and Q_{heater} is the heat generated in the heater cells based on the measured electrical current and heater voltage. ΔP is defined as the pressure difference between the inlet and outlet of the cooler. $\dot{V}$ represents the volumetric flow rate. The thermal resistance R_{th} in this work also includes the heat losses through the cooler material into the ambient, and also the heat losses through the bottom side of the assembled test board. Therefore, the definitions of normalized thermal resistance R_{th}^* and normalized pump power W_p^* are respectively defined as:

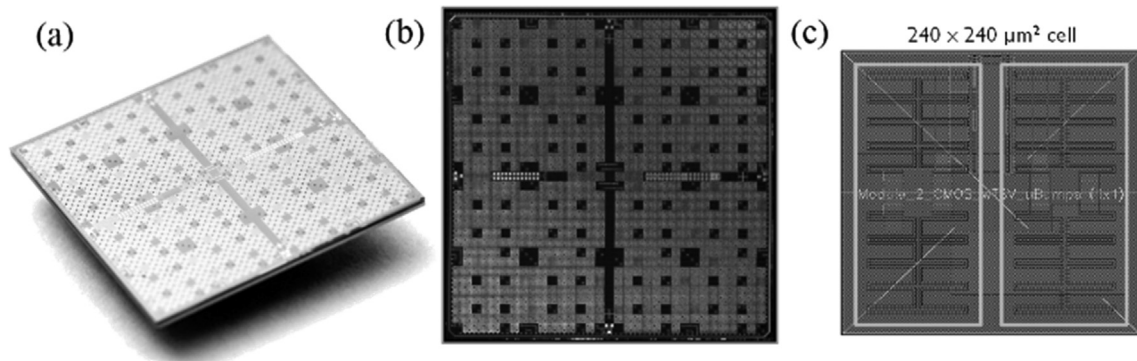


Fig. 3. Details of the single PTCQ thermal test chip: (a) overview of single thermal test die; (b) layout of the thermal test die; (c) heater cell overview.

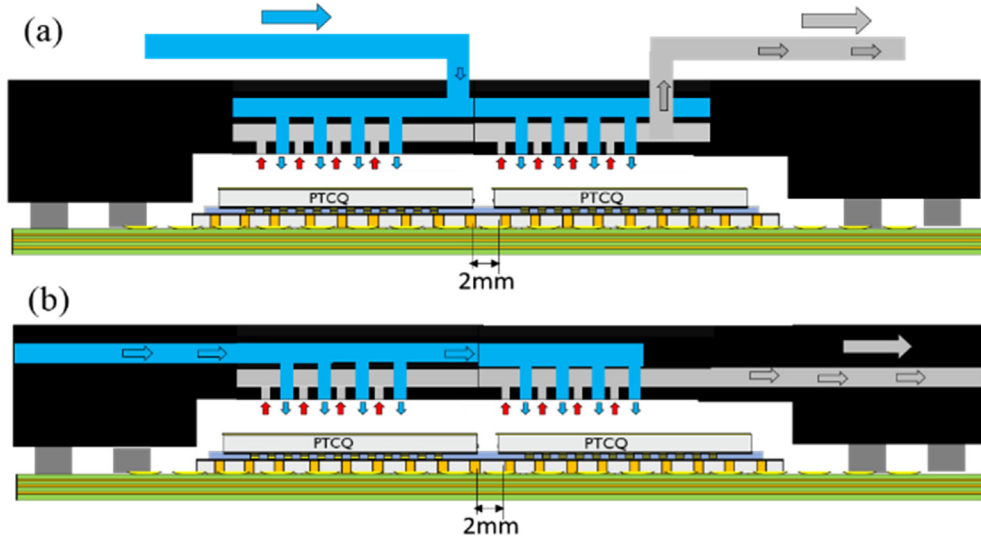


Fig. 4. Cooler schematics for the dual-chip module: (a) vertical feeding design; b) lateral feeding design.

$$R_{th}^* = \frac{(T_{avg} - T_{in}) \cdot A}{Q_{heater}} \quad (3)$$

$$V^* = \dot{V}/A \quad (4)$$

$$W_p^* = W_p/A \quad (5)$$

The normalized thermal performance represents the intrinsic thermal performance, independent of the chip and package area. This means that the thermal performance of the dual-chip package cooler in terms of absolute thermal resistance R_{th} can be predicted based on the results from the single chip cooler [30] if the same cooling cell dimensions are used (nozzle diameter and pitch, nozzle plate thickness). This normalization concept for the thermal resistance based on the chip area has been introduced in literature, to compare the thermal performance of different cold plates, with different sizes of heat sources at different flow rates [33–35]. In our design with the dual-chip cooler, the same nozzle array geometry properties are used, and the normalization concept is experimentally validated in Section 4.2.

Moreover, to generalize the results, the overall thermal performance is also defined as the dimensionless Nusselt number Nu_{avg} . They are primarily a function of the Reynolds number Re_d :

$$Nu_{avg} = \frac{h_{avg} \cdot d_i}{k_f} \quad (6)$$

$$h_{avg} = \frac{Q_{heater}}{(T_{avg} - T_{in}) \cdot A} \quad (7)$$

where Nu_{avg} is the averaged Nusselt number, h_{avg} is defined as the average heat transfer coefficient and k_f is the thermal conductivity of the fluid. The correlation between the Nu and Re will be shown in Section 5.2.

$$Re_d = \frac{\rho \cdot d_i \cdot V_{in}}{\mu} \quad (8)$$

where the Re_d is defined as the Reynolds number based on the jet inlet nozzle diameter, and the V_{in} is defined as the averaged inlet nozzle velocity, d_i is the inlet nozzle diameter, ρ is the density of the fluid, and μ is the dynamic viscosity of the fluid. Moreover, the characteristic length scale used in the full cooler level model is based on the jet inlet nozzle diameter d_i , rather than the inlet diameter of the global collector of the cooler.

2.3. 3D printed dual-chip package cooler design

For the cooler design of the dual-chip package, the objectives are listed as below:

- Targeted cooling for both chips;
- High cooling performance and low pressure drop;
- Small form factor of the cooler;

Moreover, the design constraints such as flow loop connections, cooler size limitations, assembly constraints and the manufacturing capabilities should be taken into account. Thanks to the recent advancements in additive manufacturing, this fabrication method becomes an interesting fabrication option for these coolers with typical nozzle diameters of several 100 μm . Additive manufacturing, or 3D printing enables to use low cost materials for the cooler fabrication, to print the whole geometry in one piece, to customize the design to match the nozzle array to the chip power map and to fabricate very complex internal structures. This last feature allows the design of complex cooler cavities that cannot be fabricated with convective fabrication techniques. Moreover, the inlet and outlet divider structures can be printed as hollow cylinders, which can significantly reduce the pressure drop compared to square shape dividers and reduce the number of layers required in the cooler design.

The schematic concepts of the direct jet impingement cooler for the dual-chip module are shown in Fig. 4, containing four main parts: inlet plenum, outlet plenum, nozzle plate and impingement cavity. Fig. 4(a) shows the reference cooler design with vertical coolant supply connectors. The details of the main parts in the cooler structure of vertical feeding design are indicated in Fig. 5. This design is an extension from single die cooler shown in Fig. 1. Since the optimized geometry parameters for the jet nozzle array of the single die cooler are already investigated in our previous study [28], the design of the package level cooler for the dual-chip module will use the same nozzle array (4 × 4 inlet nozzle array and 5 × 5 outlet nozzle array with 600 μm diameter) as for the single chip package cooler, targeted at each of the chips in the module. Fig. 5 shows the geometry information tabulated in Table 1. Table 1 lists the geometry comparison between the single chip cooler and the dual-chip module cooler. The cooler is designed to match to the package area of 35 × 35 mm². The cavity height of the cooler (i.e. the distance between the nozzle plate and the chip surface) is 0.6 mm. The cooler is placed over the stacked chips on the interposer, as shown schematically in Fig. 6, which also indicates the position of the sealing rings between the cooler and the package substrate. Moreover, the O-

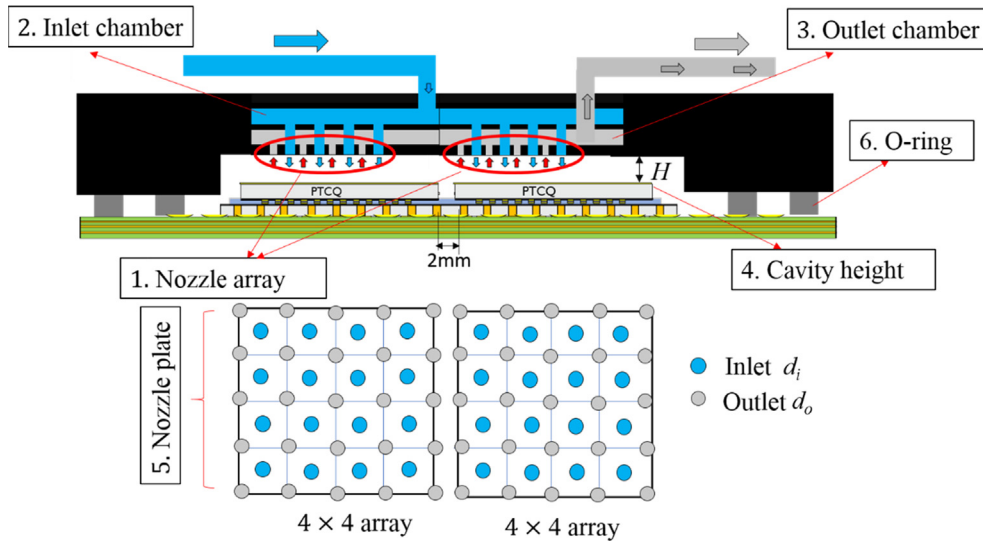


Fig. 5. Cooler geometry parameters for vertical feeding design: 1-nozzle array; 2-inlet chamber; 3-Outlet chamber; 4-Cavity height; 5-Nozzle plate; 6-O-ring.

Table 1

Geometry comparison between single die cooler and dual-chip module cooler.

Geometry		Single chip cooler	Dual-chip cooler
Nozzle array	N	4×4	4×4 per die
Inlet chamber height		2.5 mm	2.5 mm
Inlet diameter	d_i	0.6 mm	0.6 mm
Outlet diameter	d_o	0.6 mm	0.6 mm
Cavity height	H	0.6 mm	0.6 mm
Nozzle plate thickness	t	0.55 mm	0.55 mm
Cooler size	x, y, z	$14 \times 14 \times 8.7$ (mm ³)	$35 \times 35 \times 12.6$ (mm ³)

ring can also act as a buffer for the mechanical assembly of the cooler, especially for large die packages to compensate for the potential warpage of the assembly.

As an alternative design, the lateral feeding design is introduced taking full advantage of the additional design options enabled by 3D printing. As shown in Fig. 4(b), the inlet coolant flow enters the cooler at one side and spreads in the inlet plenum to be distributed over all the nozzles of both chips. This design allows to improve the flow uniformity over the nozzles, to reduce the pressure drop in the cooler through an optimized internal design and to reduce the overall cooler thickness significantly.

Fig. 7 shows the cross-sections of the inside delivery manifold for the 2 cooler configurations. In the case of the vertical feeding configuration with an inlet and an outlet plenum above each other (Fig. 7a and b), the overall cooler thickness is 12.6 mm including the tube connection. In the case of the lateral feeding configuration (Fig. 7d), the

cooler thickness can be significantly reduced since the two plenums can be integrated on the same level. The enlarged view with the details of nozzle arrays and inlet chamber is shown in Fig. 7(e). The overall cooler thickness for this configuration is 6.6 mm, which realizes a reduction of the cooler thickness by a factor of two compared to the standard cooler configuration. The experimental and numerical comparison of the thermal and hydraulic performance of both cooler concepts will be discussed in Section 5.

2.4. Full cooler level model

In order to investigate the thermal and hydraulic performance of the coolers in more detail, conjugate heat transfer and fluid flow simulations are used in this study. The numerical simulations of the full cooler model are based on the commercial software ANSYS Fluent 18.0. The material used for the solid domain is silicon and the water used for the fluid domain. The physical property of the materials used in the numerical simulation are listed in Table 2.

The meshing details of the CFD models for both cooler designs are shown in Fig. 8(a) and (b), containing typically 5 million elements. The first layer thickness of the boundary layers is 1 μ m, which is calculated from the Y plus number ($Y^+ < 1$). The maximum boundary layer number is set as 10 with a growth rate of 1.1. The mesh independency of the simulation results has been assessed using the Richardson extrapolation resulting in a truncation error for the chip temperature in the stagnation point of 0.3%. Moreover, the turbulent model used in the CFD modeling is transition SST model [36], which can capture the flow behaviors between the laminar flow and turbulent flow. The “SIMPLE”

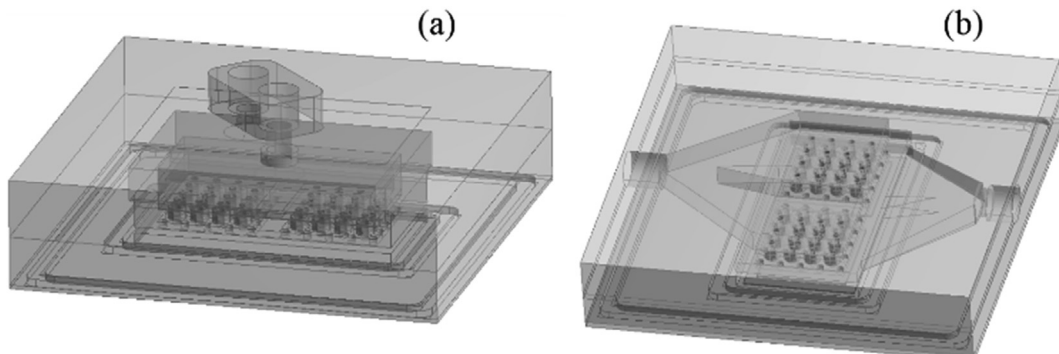


Fig. 6. CAD design of the two different coolers: (a) vertical feeding design; (b) lateral feeding design.

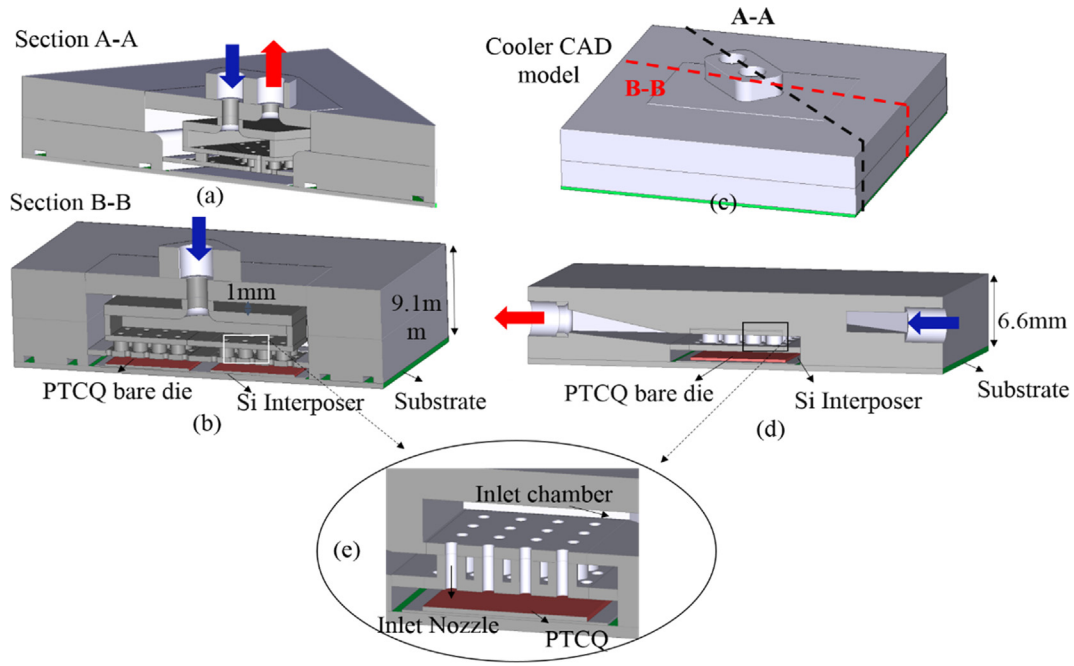


Fig. 7. Cross-section schematics of the two impingement jet cooler configurations: (a) vertical feeding design with cross section A-A; (b) vertical feeding design with cross section B-B; (c) full CAD model of vertical feeding design; (d) cross section of lateral feeding design; (e) enlarged view of the nozzle array and inlet chamber.

Table 2

The physical property parameters of materials used in CFD simulation.

Material	Density	Specific heat	Thermal conductivity	Viscosity	Temperature
Unit	kg/m ³	J/(kg K)	W/(m K)	Kg/(m s)	°C
Silicon	2329	556	149	0.1	–
Water-liquid	999.7	4197	0.6	0.0013	10

algorithm is used as the solution method. The QUICK scheme is selected as the numerical scheme. The convergence criteria were set at $1e^{-5}$ for continuity, $1e^{-6}$ for energy and $1e^{-6}$ for k , ω and momentum (x , y and z components of velocity), respectively.

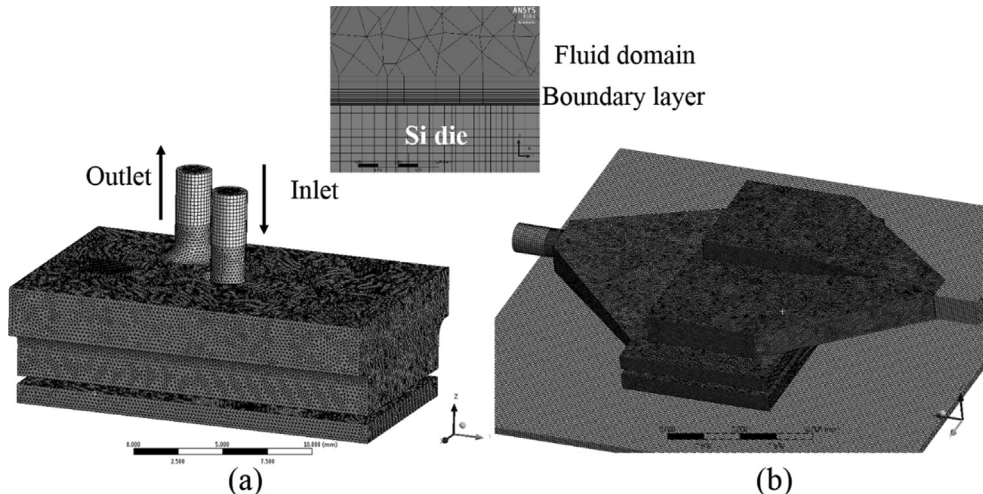
For the boundary conditions of the CFD model, uniform power dissipation is applied as a constant heat flux in the active die while there is no power in the passive die. The bottom part under the active die and bottom die, such as the Cu pillars and underfill material, the package substrate, the solder balls and the PCB are neglected, based on

our previous study [32]. The ambient temperature is considered to be at 25 °C. The inlet temperature for the CFD model is set to 10 °C. The flow boundary condition for the inlet is based on the velocity inlet with a specified constant velocity value across the inlet area. The boundary condition for the outlet is set as ‘pressure out’ ($P_{out} = 0$). For all the simulations, the net imbalance of overall mass, momentum and energy is kept below 0.02%. The CFD models for the bare die package cooling will be used in Section 5.2 for the thermal and hydraulic performance comparison of the reference cooler design and the improved design.

3. Cooler assembly and experimental set-up

3.1. Fabricated cooler assembly

The cooler is printed as a single part by stereolithography (SLA), layer by layer by curing the photosensitive polymer material with exposure of a UV light source. The chosen polymer material, Somos WaterShed XC is a water-resistant material, which shows ABS-like



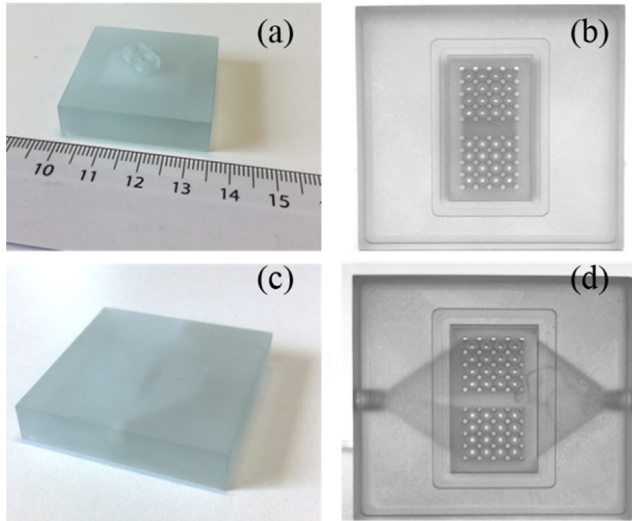


Fig. 9. Dual-chip module cooler demonstrators: sideview (a) and bottom view (b) of vertical feeding cooler; sideview (c) and bottom view (d) of lateral feeding cooler.

properties and good temperature resistance. The heat deflection temperature (HDT) of ‘Somos WaterShed XC’ is around 60 °C [37]. Therefore, the temperature of the coolant should be below 60 °C to remain in the safe temperature range for the cooler material. The printed coolers are shown in Fig. 9. The measured average nozzle diameter is $570 \mu\text{m} \pm 20 \mu\text{m}$ which only deviates 5% from the nominal design values of $600 \mu\text{m}$. The uncured resin in the cavity needs be removed using a chemical solvent after all the parts are finished. Since the cooler is printed as a single part, there is no visual access to check for internal defects or blocked nozzles. Therefore, the Scanning Acoustic Microscopy technique (SAM) is introduced in our research to evaluate the cooler quality [30]. The cooler is finally assembled onto the thermal test board. The final process flow of the dual-chip module cooler with lateral feeding design is shown in Fig. 10.

3.2. Experimental set-up

Fig. 11 shows the experimental set-up for the two different coolers, where the assembled dual-chip module cooler packages are placed in a socket to perform the thermal measurements. A known amount of power is generated in a chosen distribution, in this case all the heating elements on the chip are activated, while the full chip area temperature distribution is measured in the diodes of the PTCQ test chips. The temperatures are reported as temperature differences with respect to

the coolant inlet temperature. The coolant used in this study is DI (deionized water) water. The cooler is connected in a closed loop, including the pump, flow, pressure and temperature sensors, filters and a heat exchanger. All the sensors in the set-up are connected and controlled by LabView, allowing operation of the flow loop either in a controlled mass flow rate mode or a controlled pressure mode [32]. The accuracy of the mini Cori flow meter is $\pm 0.2\%$ RD (percentage of reading). A differential pressure gauge that can withstand a static pressure of 10 bar and measure a pressure drop between 0.2 bar and 5 bar. The accuracy is $< \pm 0.5\%$ FS (full scale). A stainless-steel basket filter is used for the pump with a screen of $25 \mu\text{m}$. Thermocouples are used to measure the outlet temperature with an accuracy of ± 0.3 °C. For the accuracy of the thermal test chip, the calibrated sensitivity of the temperature sensor on the test chip is $-1.55 \pm 0.02 \text{ mV}/^\circ\text{C}$ for a current of $5 \mu\text{A}$ in the temperature range between 10 and 75 °C with 95% confidence interval. The propagated measurement uncertainty results in a value of $\pm 1.8\%$ for the reported thermal resistance measurements [32]. Moreover, the uncertainty of the Nusselt number Nu_{avg} shown in Eq. (6) is $\pm 5.3\%$. The calculation of this measurement uncertainty is based on the propagation of the of the uncertainties of the measured chip power ($\pm 0.1\%$), the average chip temperature measurement ($\pm 1.5\%$), inlet temperature measurement ($\pm 1\%$), and the measured nozzle diameter ($\pm 3.5\%$). The thermal performance estimation of the assembled cooling solution also includes the heat losses through the cooler material into the ambient and the heat losses through the bottom side of the assembly, through the test board.

4. Characterization of standard cooler

4.1. Reference temperature measurement

One of the objective of the study is to compare the effect of the impingement cooling solution on bare die and lidded packages in terms of the thermal resistance of the heated chip and the thermal coupling between the two chips in the package. Since the package type (with lid and without lid) is different, this might have an impact of the heat conduction inside the chip package and consequently on the thermal resistance and the thermal coupling. Therefore, the packages are first measured without any active liquid cooling applied. There are two main thermal paths towards the ambient for the heat generated in a chip in the package: one is from the top side of the package, through the cooler (R_{cooler}); the second parallel thermal path is downward through the package substrate and PCB (R_{bottom}). The overall thermal resistance is the parallel connection between the two thermal paths. In the first reference measurements, the top side of the package is insulated, and the heat is removed through the bottom part of the package, enabling the characterization of the bottom thermal part, as shown in Fig. 12(a). The thermal resistance network is shown in Fig. 12(b) for the

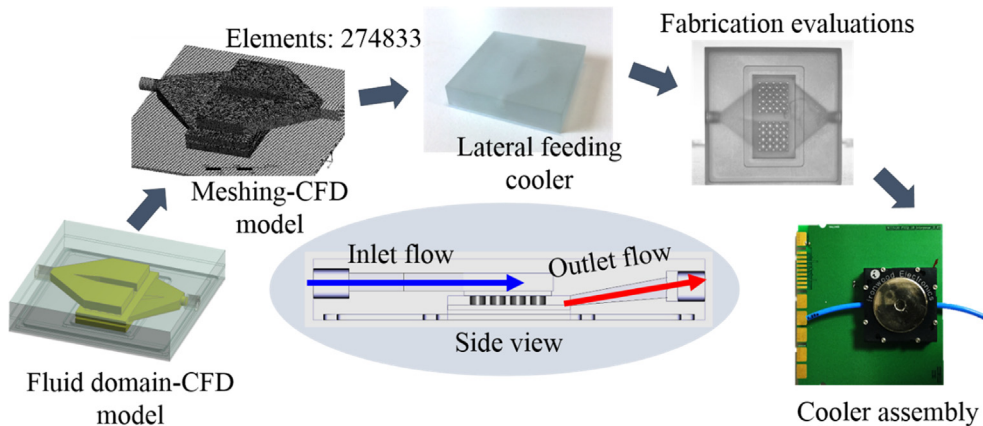


Fig. 10. Design flow for the dual-chip module cooler: lateral feeding design.

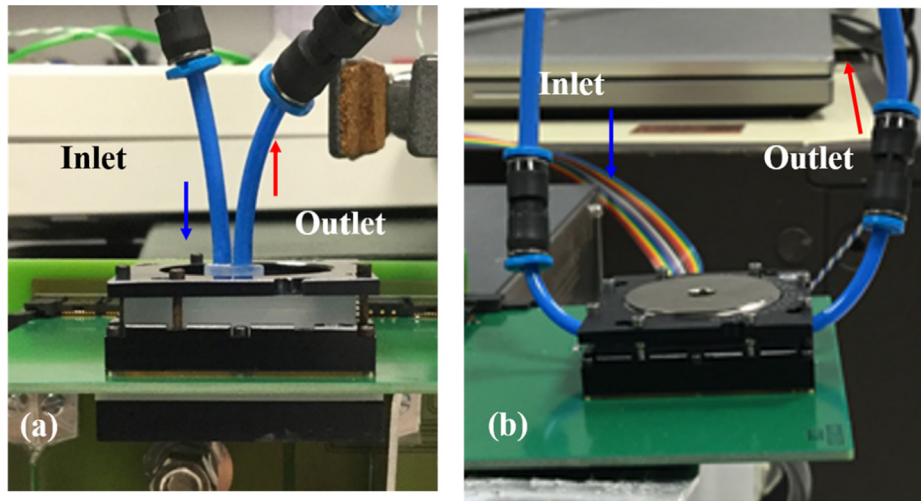


Fig. 11. Experimental set-up for (a) vertical and (b) lateral feeding cooler.

illustration of the heat flow simplified in thermal paths. Fig. 12(c) shows the measured temperature distribution in the test chip for a power dissipation of 1.7 W. This corresponds to an average thermal resistance of 15.1 K/W, or an average area-normalized thermal resistance of $9.6 \text{ cm}^2 \text{ K/W}$ for the bottom thermal path through the package substrate and PCB.

4.2. Bare die versus lidded package cooling

For the thermal measurements with the applied cooling, 50 W power is dissipated in each chip of the dual-chip package, while the full temperature map of the chips is measured for a flow rate of 1000 ml/min (for cooler, thus 500 ml/min per chip). The thermal performance of the 3D printed cooler is compared for the lidded and the lidless

packages in this section. The measured chip temperature distribution maps for both packages are shown in Fig. 13. The temperature profile at the center of the chips is shown in Fig. 14 to allow a more detailed comparison of the thermal behavior. The comparison of the temperature profiles of the two package reveals a significant difference for both the heated chips. The overall thermal resistance of the logic chip is a factor of 2 to 3 higher in case of the lidded package compared to the lidless package. This large difference is mainly caused by the presence of the thermal interface material.

The measurement results for the lidded and lidless packages are summarized in Table 3 for different flow rates. The presence of the lid (and mainly the TIM) results in a higher chip temperature, where the relative impact of the lid increases as the flow rate increases since the convective thermal resistance decreases with the flow rate. The

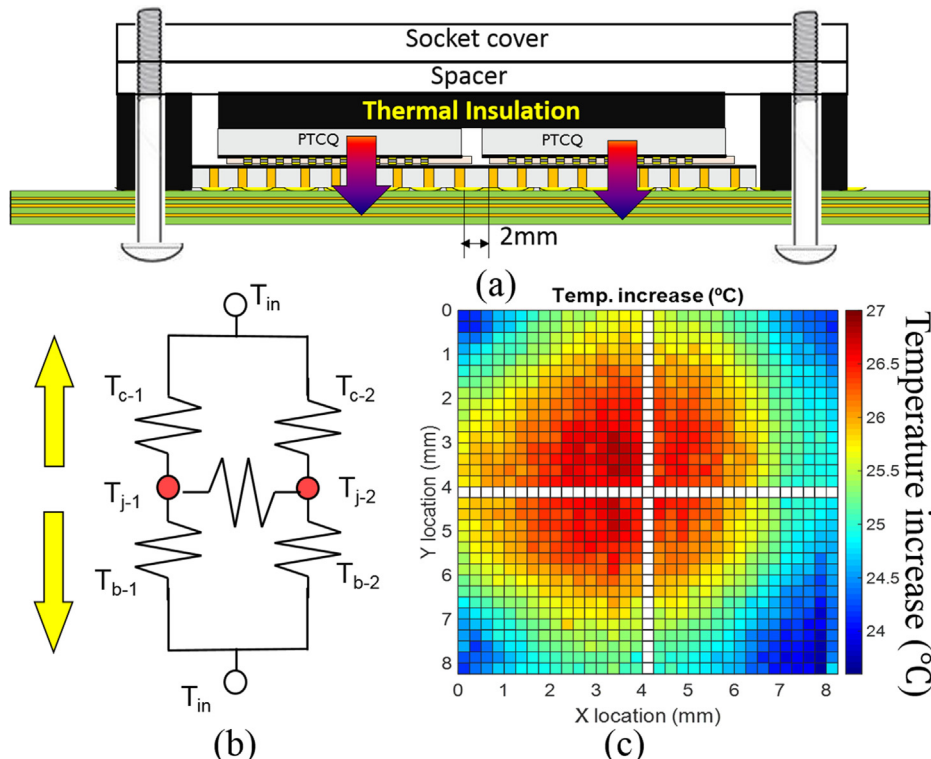


Fig. 12. Reference temperature measurement with thermal insulation: (a) schematic view of the measurement setup with thermal isolation on top of the die surface; (b) thermal resistance network of the thermal path of the interposer package; (c) temperature increase map of the interposer package with thermal insulation.

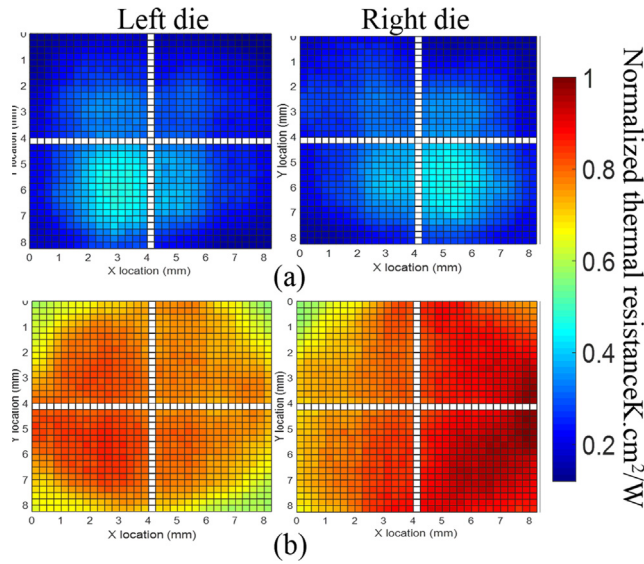


Fig. 13. Temperature distribution on two heated die with (a) lidless cooling and (b) lidded package cooling (left chip power = 50 W, right chip power = 50 W, flow rate = 1000 ml/min, vertical feeding cooler).

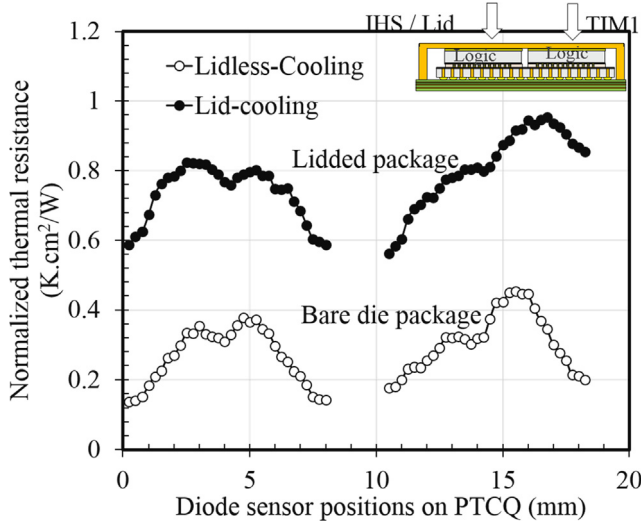


Fig. 14. Thermal measurement comparison between the lidless package cooling and lidded package cooling (single chip power = 50 W, flow rate = 1000 ml/min, vertical feeding cooler).

Table 3

Thermal comparison between the lidded and lidless packages.

Flow rate (ml/min)	Average thermal resistance-Total (cm² K/W)		Average thermal resistance of cooler (cm² K/W)		Relative heat loss	
	Lidded	Lidless	Lidded	Lidless	Lidded	Lidless
300	0.85	0.47	0.93	0.49	9.71%	5.15%
400	0.80	0.41	0.87	0.43	9.09%	4.46%
600	0.75	0.33	0.81	0.34	8.47%	3.56%
1000	0.68	0.26	0.73	0.27	7.62%	2.78%

additional thermal resistance of the TIM and lid can be estimated as $0.45 \text{ cm}^2 \text{ K/W}$. The thermal conductivity of the TIM is calibrated as 1.9 W/m K and a targeted thickness of $80 \mu\text{m}$. The reported overall thermal resistance values are the result of the combined heat removal through the cooling solution and the heat losses through the package.

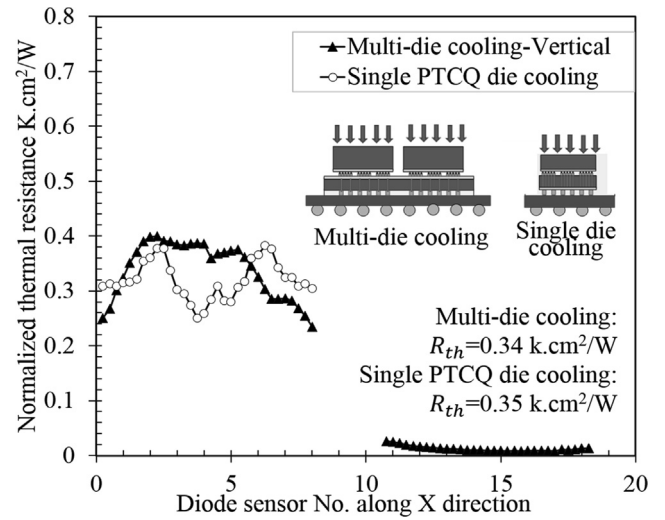


Fig. 15. Experimental comparison with dual-chip cooling and single chip cooling at flow rate of 600 ml/min (Vertical feeding manifold).

By combining the results with the reference measurements, the heat losses through the package can be estimated. The relative values for the heat losses are shown in Table 3 for the two packages for different coolant flow rate values. These results show that the heat losses are limited to values from 2% for a high flow rate to 5% for a low coolant flow rate and therefore the majority of the heat is removed through the cooling solution on top of the package. Table 3 also shows the thermal resistance values for the top heat flow part only, after correction for the heat losses. Since the heat losses through the bottom package are small, the type of packaging does not have a significant impact on the thermal resistance values and the thermal coupling in the packages.

4.3. Comparison with single die

The dual-chip module cooler has been characterized for an overall flow rate of 600 ml/min (divided over both chips in the module) and for a power dissipation of 50 W in the left chip and no power dissipation in the right chip. The normalized thermal resistance for vertical feeding of the cooler is shown in Fig. 15 for a horizontal profile across the two chips. The cooling performance for the single chip cooler [28] with the same nozzle array and the same normalized flow rate, in this case of 300 ml/min per chip, shown in the same figure, results in a similar value of $0.35 \text{ cm}^2 \text{ K/W}$. This comparison shows that the average normalized thermal resistance can be used to extrapolate the thermal performance for different chip sizes or multiple chips, supporting the approach of a scalable cooling solution with a constant intrinsic thermal performance for the unit cooling cell with constant flow rate per unit cell. Therefore, this scalable approach can be used to design specific nozzle arrays for different chip sizes in the module, as discussed in Section 2.1. It should be noted that there is a discrepancy between the single chip and dual-chip cooling for the actual temperature profile. These local differences are due to the designs of the inlet/outlet chamber in the plenum level resulting in different local flow distribution.

4.4. Thermal coupling effects

In the next step, the thermal coupling between the active die and passive die in the dual-chip module cooled by the package level impingement cooler is investigated. Therefore, the power dissipation in the left chip referred as “logic die” is set as 50 W while no power dissipation is applied in the right chip referred as “Memory die”. The temperature distribution map of the bare die cooling with vertical feeding cooler and of the lidded package cooling are shown in Fig. 16(a)

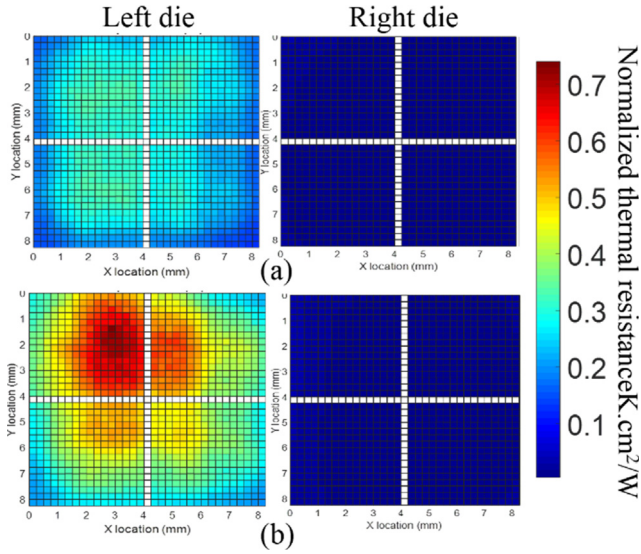


Fig. 16. Normalized thermal resistance measurements ($\text{cm}^2 \text{K/W}$) for the (a) lidless cooling and (b) lidded package cooling on the interposer package (active die = 50 W, passive die = 0 W, flow rate = 1000 ml/min).

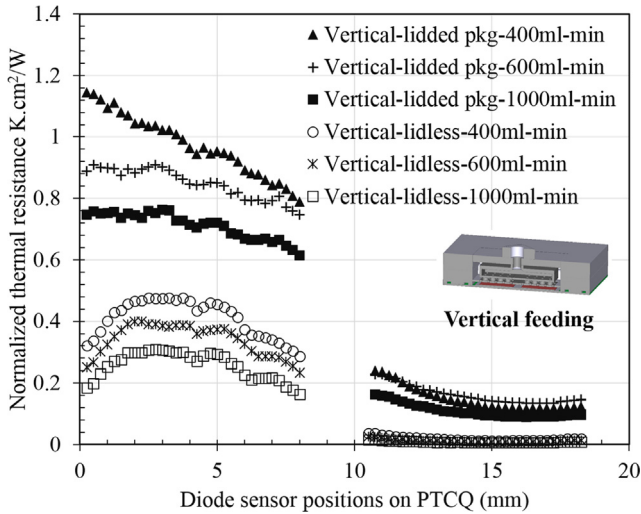


Fig. 17. Normalized thermal resistance ($\text{cm}^2 \text{K/W}$) measurements profile comparison between lidless cooling and lidded package cooling under different flow rate (active die = 50 W, passive die = 0 W, flow rate = 1000 ml/min).

and (b) respectively for an overall flow rate of 1000 ml/min. The temperature profiles for three different flow rates is plotted in Fig. 17. It can be observed that the thermal coupling between the active die and the passive die is much higher in the lidded package compared to the lidless package, due to the heat spreading in the Cu lid. In the bare die package, the temperature increase of the passive die is very limited due to the absence of thermal coupling path of the lid. In the lidded package, passive die temperature is much higher and also shows a clear temperature gradient from the left side to the right side of the chip.

The thermal coupling in a multi-die module can be expressed as a part of the thermal resistance matrix R , which is used for multi-chip modules describing the thermal interactions between the different heat sources [38–40]:

$$\begin{bmatrix} T_{\text{left}} \\ T_{\text{right}} \end{bmatrix} \cong \begin{bmatrix} R_{\text{left, left}} & R_{\text{left, right}} \\ R_{\text{right, left}} & R_{\text{right, right}} \end{bmatrix} \cdot \begin{bmatrix} P_{\text{left}} \\ P_{\text{right}} \end{bmatrix} + \begin{bmatrix} T_{\text{in}} \\ T_{\text{in}} \end{bmatrix} \quad (9)$$

The resistance R_{ij} in the matrix of (5) is the temperature rise of heat source 'i' above the ambient temperature (inlet temperature), caused by

Table 4

Thermal coupling at different flow rate.

Flow rate (ml/min)	Lidded pkg	Lidless pkg
300	15.7%	7.1%
400	15.98%	5.3%
600	17.38%	4.6%
1000	26.1%	3.4%

unit heat dissipation of source "j":

$$R_{ij} = \frac{T_i - T_{\text{in}}}{P_j} \quad (10)$$

For a dual-die package, 4 thermal resistance terms are required to describe the thermal resistance matrix: the self-heating thermal resistance terms on the diagonal and thermal coupling resistance terms, or mutual heating effects on the cross-diagonal. The thermal resistance (6) should be obtained in case of uniform power dissipation in one of the chips while there is no power dissipation in the other chip(s) in the 3D package. In case of non-uniform power dissipation, the concept of thermal resistance is not meaningful. Therefore, the thermal coupling between the passive die and active die in this study can be expressed as below based on the average chip temperatures:

$$R_{\text{coupling}} = \frac{T_{\text{passive}} - T_{\text{in}}}{T_{\text{active}} - T_{\text{in}}} \quad (11)$$

Table 4 shows the thermal coupling effects for lidless cooling and lidded package cooling at different flow rate. Moreover, the normalized thermal resistance and thermal coupling as function of the flow rate for the lidded package cooling and lidless cooling are plotted in Fig. 18. As shown in Fig. 18 at a flow rate of 1000 ml/min, the thermal coupling is 7.7 times higher in the lidded package, compared to the lidless package. For the lidless package cooling, it can be seen that the thermal coupling reduces for increasing flow rate. This is because the main thermal path is dominated by the high cooling efficiency of microjet cooling without TIM and lid. While for lidded package cooling, the thermal coupling effects become higher with increasing flow rate. The reason for this is that the relative contribution of the conductive thermal resistance from the TIM and lid increases, as the convective resistance decreases. Typical test case values for lidded MCP with heat sink presented in literature [41], report a thermal coupling value of 31.2%. The presented bare die cooling with a flow rate of 1000 ml/min achieves a 9 times reduction of the thermal coupling.

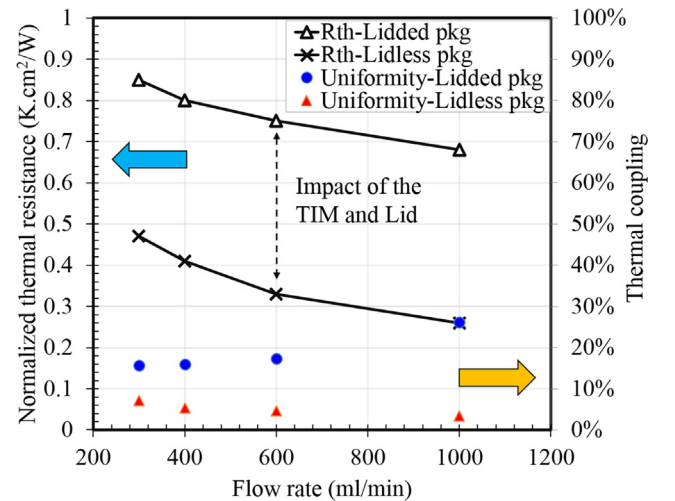


Fig. 18. Normalized thermal resistance and thermal coupling as function of the total flow rate for lidded package and lidless package.

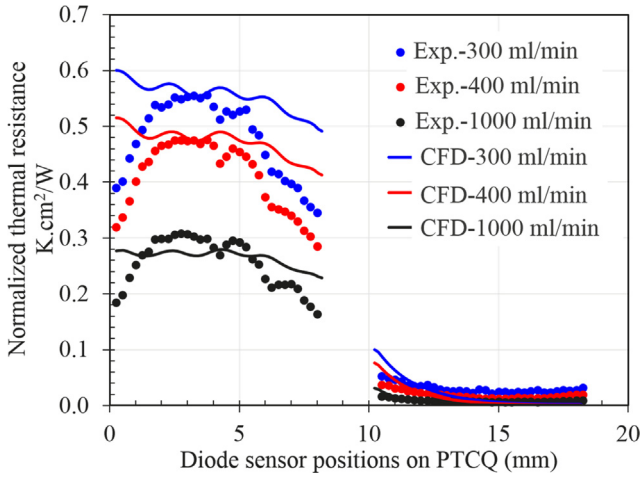


Fig. 19. Experimental validation of CFD modeling for bare die cooling under different flow rates (logic power = 50 W; memory power = 0 W).

4.5. Modeling validation

In this section, the comparison between the CFD modeling results and experimental results is discussed and analyzed. As shown in Fig. 19, full cooler level CFD modeling results with the temperature profile across the two thermal test die source regions show the similar trend with the experimental data under different flow rate. The lower temperature around the chip edge in the experiments is due to the full submerged of the thermal test die inside the liquid. An additional aspect of the discrepancy between the experimental and modeling results, is the different coolant impact for impingement cooling on the die surface only or also on the chip sides. It is shown in literature [42,43], that jet impingement hybrid body cooling with a submerged die has better cooling performance than jet impingement surface cooling on the die surface only. This is because the hybrid body cooling can provide extra cooling for the chip by channel cooling to the side surfaces, resulting in a lower temperature at the chip edges. In Fig. 20, the thermal resistance distribution maps are compared for the experiments and the CFD model results for the lidless interposer package. The nozzle cooling patterns can be clearly distinguished from the modeled temperature distribution that assumes 100% uniform heat dissipation while the actual PTCQ power map is quasi-uniform with 75% heater uniformity. The measured

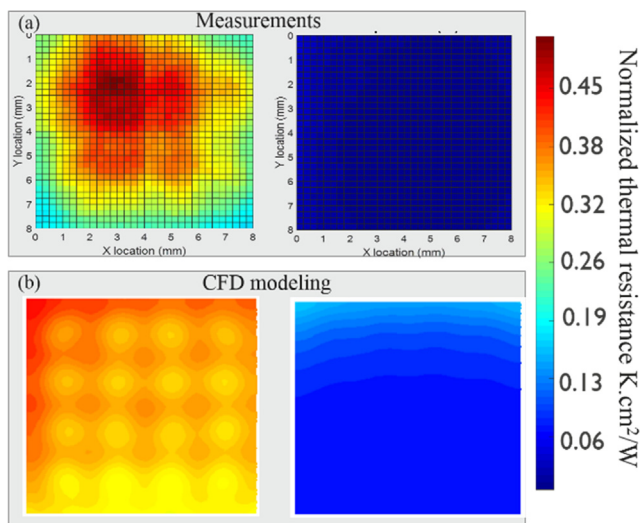


Fig. 20. Normalized thermal resistance ($\text{cm}^2 \text{ K/W}$) distribution comparisons between the measurements and CFD modeling for bare die liquid cooling (logic power = 50 W; memory power = 0 W; flow rate = 300 ml/min).

averaged temperature for the “logic” die based on the bare die cooling at flow rate of 300 ml/min is $0.47 \text{ cm}^2 \text{ K/W}$ while the modeling averaged temperature is $0.46 \text{ cm}^2 \text{ K/W}$. In general, the full cooler level CFD modeling results agrees well with the measurement data with respect to the average temperature, however differences in local temperature distribution become visible at the location of non-heated parts due to the high heat removal rate. For this level of cooling, more details of the chip power map should be included in order to predict the detailed chip temperature map. Lower temperature around the chip edge in the experiments can be explained due to the absence of the heaters there. The difference between the CFD model and the experimental data for the average chip temperature is 12.6% at a flow rate of 300 ml/min and only 2% at a flow rate of 1000 ml/min. Therefore, we use uniform heater pattern for the modeling study to save the computation cost.

5. Lateral feeding cooler performance

In this section, the thermal and hydraulic performance of the lateral feeding cooler design, introduced in Section 2.3, will be evaluated and compared with the standard vertical feeding design.

5.1. Experimental comparison

Fig. 21 shows the comparison of the thermal performance for the vertical feeding design and the lateral feeding design on both the bare die and lidded packages with a power dissipation of 50 W in one active chip and a flow rate of 1000 ml/min. Fig. 21(a) and (b) show the full temperature maps for the lidded packages while Fig. 21(c) show the temperature profiles for the measured cases. It can be seen that the temperature profiles for both coolers are very similar: the difference for the active die temperature is only 4% for both the bare die and lidded cooler. The thermal comparison between the two designs is summarized in Table 5. In general, it can be seen that the normalized thermal resistances for the vertical feeding scheme and the lateral feeding scheme

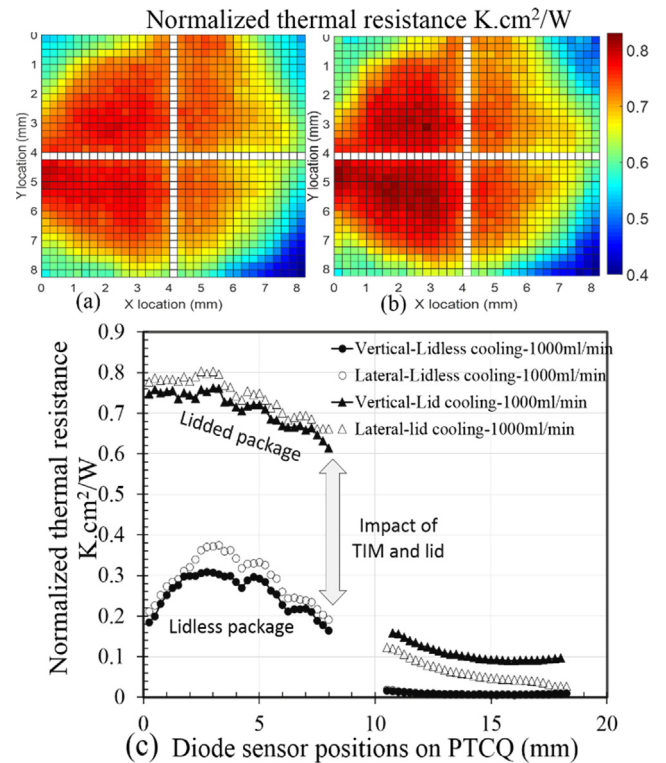


Fig. 21. Thermal measurement comparison in active die between the vertical feeding (a) and lateral feeding (b) design (logic power = 50 W, memory power = 0 W, flow rate = 1000 ml/min).

Table 5

Lidless cooling comparison between the vertical feeding and lateral feeding designed cooler at different flow rates.

Flow rate (ml/min)	Area averaged thermal resistance (cm ² K/W)		Maximum thermal resistance (cm ² K/W)	
	Vertical	Lateral	Vertical	Lateral
300	0.47	0.48	0.59	0.63
400	0.41	0.42	0.52	0.57
600	0.33	0.33	0.43	0.50
1000	0.26	0.27	0.34	0.37

are very similar to each other for all flow rates for both the lidded and bare die packages. This comparison proves that the improvement of the plenum shape to minimize the flow resistance does not interfere with the thermal performance of the cooler.

5.2. Modeling comparison

The thermal and hydraulic performance of the lateral feeding cooler can also be compared to the vertical feeding cooler using the CFD simulations introduced in Section 2.2. Moreover, these simulations can be used to assess the flow distribution in the cooler and the temperature distribution in the chip.

The first part of the comparison is the nozzle inlet velocity uniformity and pressure drop over the cooler between the vertical feeding design and the lateral feeding design. Fig. 22 shows the comparison of the velocity field inside the vertical and lateral cooler design. The flow streamlines inside the cooler are shown in Fig. 22(a) and Fig. 22(b). The cross section view of the velocity is shown in Fig. 22(c) and Fig. 22(d). For the vertical feeding scheme, the coolant is supplied in the center of the cooler. Therefore, the flow velocity will decrease as the flow goes

from the central inlet nozzles to the outer inlet nozzles. For the lateral feeding scheme, the entering flow is separated equally into two parts for the distribution of the two dies, resulting in a more uniform distribution over the nozzles. This effect is shown in Fig. 23(a). The figure compares the distribution of the average inlet nozzle velocity for both cooler designs along a cross section of the cooler. It can be seen that the velocity distribution of the lateral feeding design is much more uniform than the vertical feeding design. Since the nozzle diameter is kept as the same, therefore, the flow rate uniformity is corresponding to the nozzle velocity distribution. The analysis of the local flow rate for all inlet nozzles shows that the uniformity for the nozzle flow rate is reduced from 25% to 11% from the vertical feeding cooler to the lateral feeding cooler. Furthermore, the overall pressure drop over the cooler is much lower for the lateral feeding design, as can be seen from Fig. 23(b). The improvement of the cooler design results in a reduction of the pressure drop over the cooler of 63% and 53% at flow rates of 100 ml/min and 1000 ml/min respectively. This pressure drop reduction is caused by the improvement of the internal geometry of the plenum and the elimination of the two 90° bends for the coolant flow in the vertical feeding design. Moreover, the comparison between the CFD modeling and experimental measurement shows a good agreement for the pressure drop across the cooler for both the vertical feeding design and the lateral feeding design.

Fig. 24 shows the detailed chip temperature distribution map comparison between the vertical and lateral design. For both coolers, the inlet liquid temperature is set as 10 °C, and the chip power for the active die is set as 50 W. The simulated average thermal resistance is 0.28 cm² K/W for vertical feeding design while the lateral feeding shows 0.26 cm² K/W. It can be seen that the average chip temperature for the two cases are very similar with each other showing 7.1% difference, which corresponds well with the experimental results of the previous section. Fig. 25(a) shows the comparison between the CFD

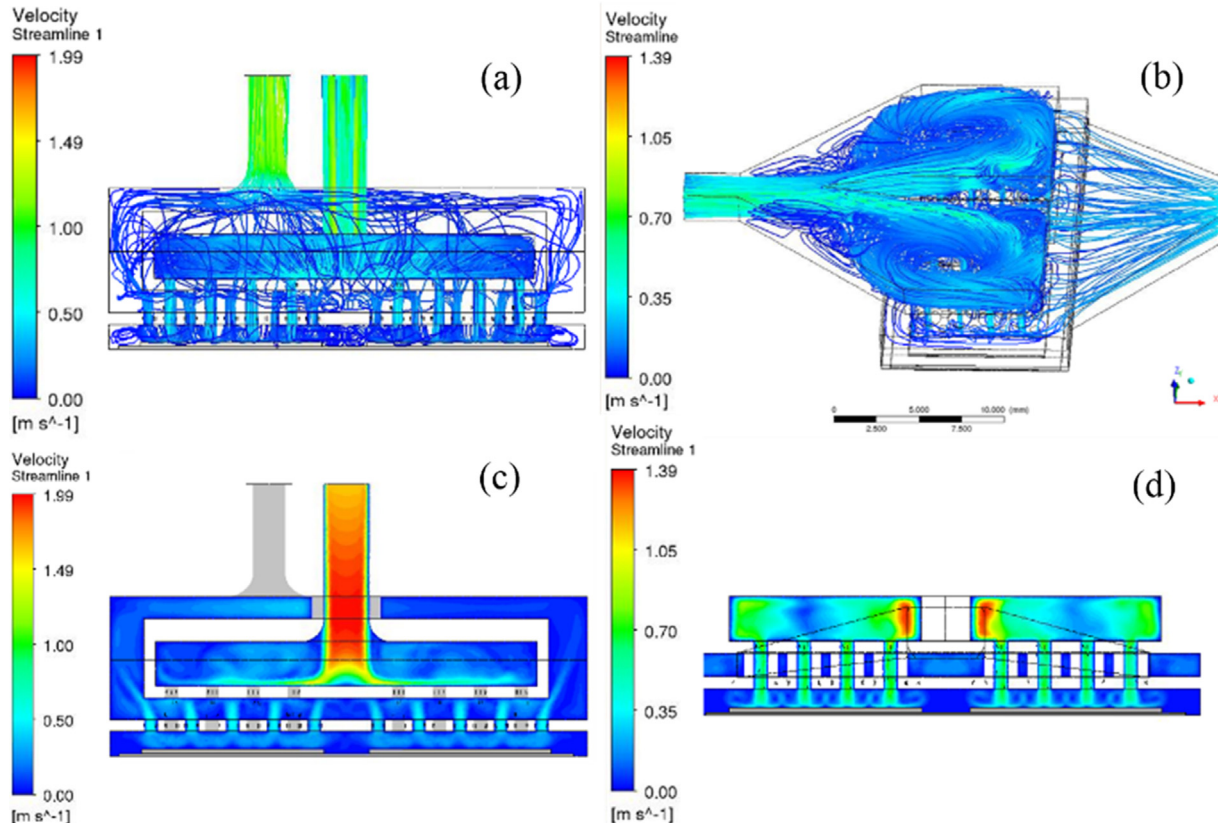


Fig. 22. CFD modeling results comparison between the vertical feeding and lateral feeding design with (a) and (b) flow streamline inside the cooler and (c) d) cross-section of the velocity field.

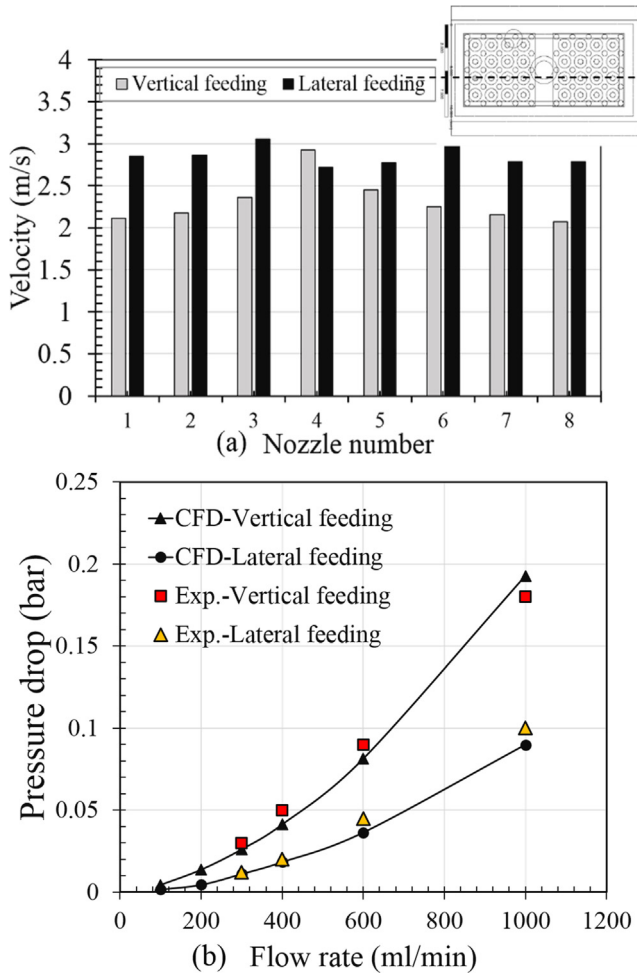


Fig. 23. CFD modeling results with (a) inlet velocity profile and (b) global level pressure drop comparison for the vertical and lateral feeding design.

modeling and experiments under different flow rate. It can be seen that the CFD modeling shows good agreement with the experimental data, especially at higher flow rate 1000 ml/min. Moreover, the thermal characteristics of the vertical feeding design and lateral design show similar behaviors. This is due to the same nozzle array design and the same power and velocity boundary condition from the system point of view. The Nusselt number and the Re were calculated based on the jet

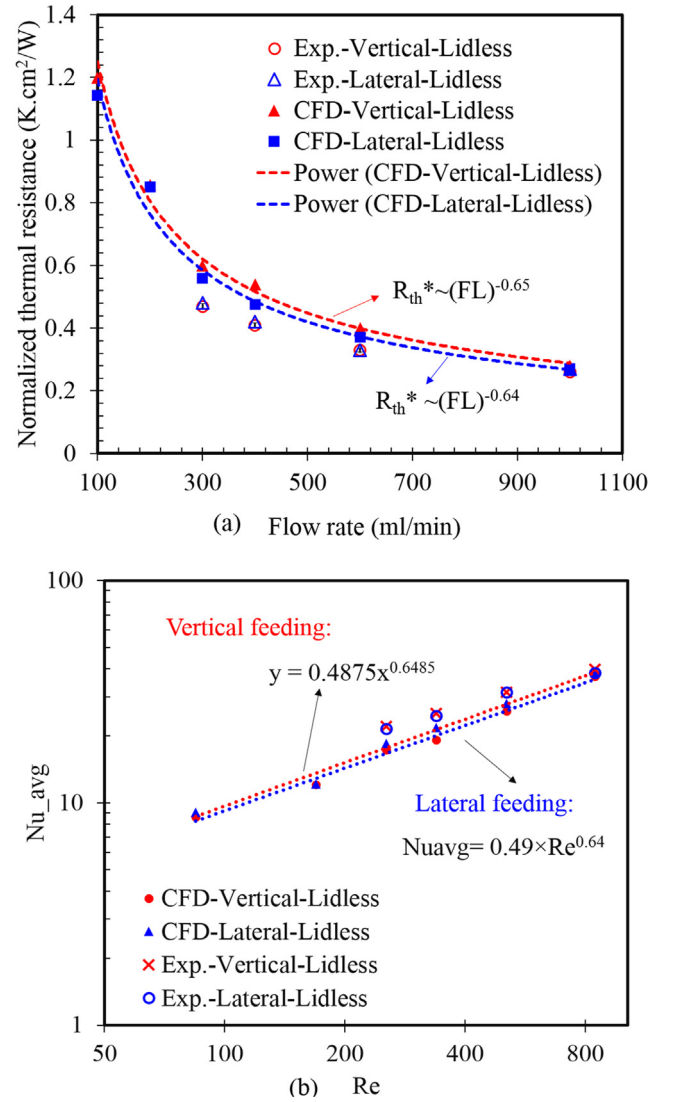


Fig. 25. Comparison of the CFD modeling results and experimental measurements between the vertical feeding and lateral feeding design: (a) Normalized thermal resistance as function of flow rate; (b) Nusselt number as function of Reynolds number.

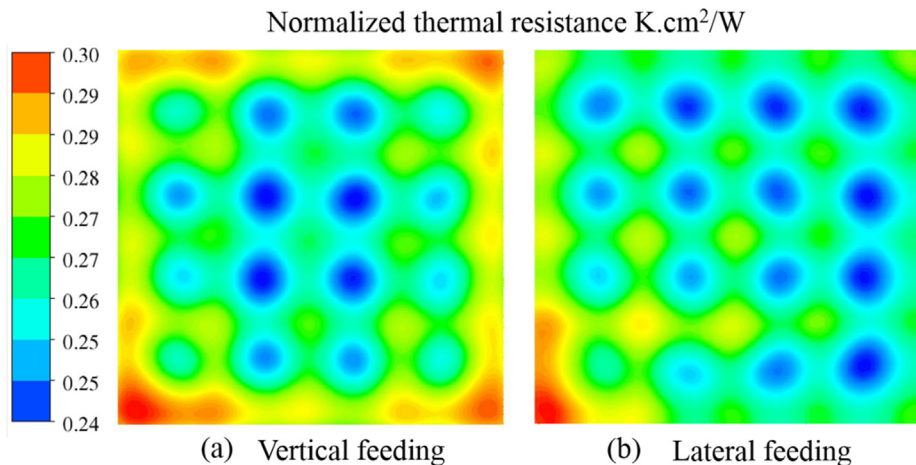


Fig. 24. CFD modeling results with chip temperature distribution comparison between the (a) vertical feeding and (b) lateral feeding design: $R_{vertical} = 0.28 \text{ cm}^2 \text{ K/W}$, $R_{lateral} = 0.26 \text{ cm}^2 \text{ K/W}$. (bare die cooling with flow rate of 1000 ml/min).

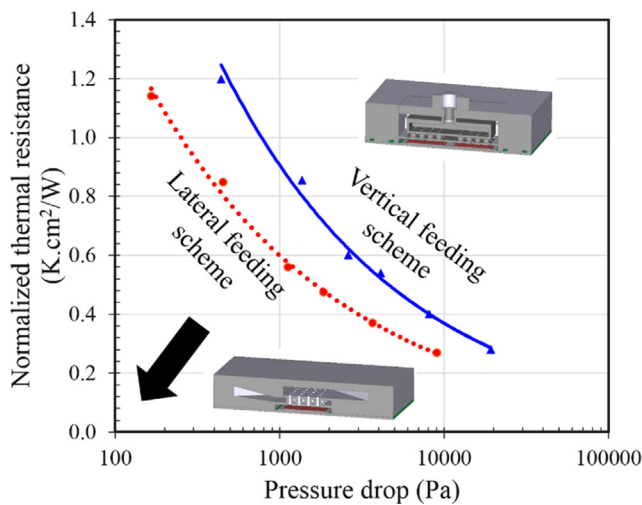


Fig. 26. CFD modeling comparison of the thermal and hydraulic performance for vertical and lateral feeding configurations (lidless cooling).

diameter shown in Fig. 25(b). The extracted Nu-Re correlations are:

$$\text{Vertical feeding design: } Nu_{\text{avg}} = 0.49 \times Re^{0.65} \quad (12)$$

$$\text{Lateral feeding design: } Nu_{\text{avg}} = 0.49 \times Re^{0.64} \quad (13)$$

The hydraulic and thermal CFD simulations for the comparison between the cooler designs on the bare die dual-chip package are summarized in Fig. 26. The chart shows the achieved normalized average thermal resistance as a function of the required pressure drop over the cooler, for flow rates ranging from 100 ml/min to 1000 ml/min. The achieved averaged thermal resistance is similar for both cooler designs at the same flow rate, however the lateral feeding cooler design requires 50–60% less pressure drop and consequently pump power to realized. At a flow rate of 500 ml/min per chip (1000 ml/min for the cooler), the normalized thermal resistance is $0.26 \text{ cm}^2 \text{ K/W}$. This means that the device temperature increase with respect to the inlet temperature would be 78°C for a heat flux of 300 W/cm^2 at a required pressure drop of 0.09 bar. Moreover, overall thickness of this lateral feeding cooler is 2 times thinner compared to the vertical feeding cooler.

6. Conclusion

In this work, we demonstrate for the first time the design, modeling, fabrication and experimental thermal and hydraulic characterization of package level 3D printed direct liquid micro-jet array impingement cooling applied to the dual-chip module used in power electronics. A scalable design methodology for the chip area is proposed and experimentally validated by comparison the thermal performance of the dual-chip package cooler with earlier single chip cooler data. The cooler has been designed for a dual-chip package that contains two advanced thermal test chips, taking advantage of the capabilities of additive manufacturing to create complex internal structures and to fabricate the cooler as a single part. The coolers, fabricated using high resolution stereolithography with the water-resistant have been assembled on bare die and lidded versions of the test vehicle.

For the bare die package, a very low thermal resistance of $0.26 \text{ cm}^2 \text{ K/W}$ is measured at a cooler flow rate of 1000 ml/min for two heated chips. The presence of the lid (and mainly the TIM) results in a higher chip temperature, where the relative impact of the lid increases as the flow rate increases. Furthermore, it is demonstrated that the bare die jet impingement cooling on the dual-chip package can realize a very low thermal coupling between the chip of only 4%, which is 9 times lower than typically reported values for multi-chip modules.

Furthermore, in this paper we introduced a novel lateral coolant feeding design for the dual-chip package cooler, which enables a reduction of the cooler height by a factor of two. The experimental comparison shows that the lateral feeding cooler achieves a very similar thermal performance as the standard vertical feeding cooler. The modeling comparison shows that the lateral feeding design can achieve a more uniform flow distribution over the nozzles in the cooler. Furthermore, this lateral feeding design can realize 50–60% reduction of the pressure and required pumping power for the same thermal performance and at the same time realize a reduction of the cooler thickness by a factor of 2 compared to the reference vertical feeding design. An optimized 3D printed fluid delivery manifold design with lateral feeding structure has a thermal resistance from junction to coolant inlet temperature of $0.26 \text{ cm}^2 \text{ K/W}$, which can cool down the heat flux up to 300 W/cm^2 for a 78°C temperature increase for a flow rate of 500 ml/min per chip and a pressure drop of 0.09 bar.

The next step is to optimize the inlet distributor to have better flow uniformity. One best option is using the topology optimization [44] design to tailor the flow for every chip module. Moreover, we can also design intermediate layer to split the flow more uniformly. Other important aspects for the future work are to address the potential reliability concerns.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2019.114535>.

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